Weekly Timeline

I. Week 1

   A. **Anika -** Construct a controlled financial environment with three assets + Estimate or calibrate asset return moments using real data.

      1. Download daily adjusted close prices for SPY, IEF, and GLD.
      2. Align the three series by common trading dates.
      3. Compute daily returns.
      4. Remove missing observations.
      5. Create one clean return matrix

      Deliverable: A clean data set of 3 aligned return series (3-5 yrs)

      1. Compute the sample mean vector
      2. Compute the sample covariance matrix
      3. Check whether the covariance matrix is valid and invertible.
      4. Save these as the fixed parameters of the controlled environment.

      Deliverable: Mean vector table, covariance matrix table

      Deliverable: Build synthetic data generation process

   B. **Tony -** Derive the true optimal portfolio under mean-variance preferences.

      1. Assume the given mean-variance investor and find closed form solution for the theoretical optimal

      Deliverable: Closed form solution

   C. **Ally -** Implement the stateless model-free Q-learning agent using only simulated returns (i.e., using the synthetic data).

      Steps: fix mu and covariance matrix

      Set a random seed for reproductivity

      Use a stateless formulation

      Define a discrete action grid over feasible portfolio weights

      Deliverables: Completed model and a list of feasible portfolio weights

   D. **Amanda -** Define the reward function

      1. For a chosen portfolio and simulated return, compute portfolio return
      2. Define the one-period reward in a mean-variance-consistent form

      Deliverable: A precise reward definition used in code and report

II. Week 2

A. **Anika, Tony** - **(EC)** Derive Optimal portfolio under mean-variance for
   1. Long-only Constraint (no shorting)
   2. Bounded weights (limited shorting/leverage)

   (by lagrangian with constraints)

Deliverables: Full benchmark derivation

Numerical benchmark weight vector w^*

Benchmark portfolio mean, variance, and utility

B. **Amanda, Ally** - Train the model to convergence

   Steps:

   Epsilon greedy exploration required

   Explain why we chose learning rate, decay, and discount factor5

   Simulate to convergence of portfolio weights (including what constitutes convergence and iteration count)

   Deliverables: Complete, Trained model

III. Week 3

A. Implement + Train the Long-only model
   1. Steps: Analogous to the week 2 description
   2. **Deliverables: Complete, Trained model**

B. Implement + Train the bounded weights model
   1. Steps: Analogous to week 2 description
   2. **Deliverables: Complete, Trained model**

IV. Week 4

A. Report Drafting/Finalization (All)

B. Compare learned portfolio policies to the theoretical benchmark.

C. Evaluate convergence speed and accuracy, including under portfolio constraints.
   1. Plot cumulative reward difference relative to $w*$, the true optimal portfolio
   2. Report number of observations needed to approach optimality

D. **Deliverables: Full Project**